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ORIGINAL ARTICLE

HRL-MOEA: a hybrid reinforcement learning-enhanced multi-objective recommendation algorithm with dynamic policy orchestration

Liang Chu[1] · Ye Tian[2]

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Abstract
Recommendation algorithms have become increasingly prevalent in modern society, addressing information overload by delivering content aligned with user preferences. While traditional approaches prioritize recommendation accuracy, singular focus on this objective often results in popularity bias. This imbalance introduces fairness concerns for item providers and detrimental feedback loops in recommendation ecosystems, highlighting the critical importance of item exposure fairness. However, balancing these dual objectives faces fundamental trade-off challenges. Existing multi-objective recommendation methods typically rely on empirically fixed genetic operators during evolutionary processes, which not only requires laborious parameter tuning but also constrains the generation of high-quality solutions. To overcome these limitations, we propose a hybrid reinforcement learning-enhanced adaptive evolutionary algorithm (HRL-MOEA). The framework synergistically integrates SARSA and Q-learning strategies through a phase-aware mechanism: during early evolutionary stages, a conservative SARSA-based self-adaptive mechanism facilitates comprehensive solution space exploration, while strategically transitioning to Q-learning's exploitation-oriented policy in later phases to accelerate convergence. This dynamic strategy is conducive to enhancing the model's evolutionary performance. Experimental results demonstrate that HRL-MOEA outperforms existing algorithms in performance effectiveness.

Keywords Multi-objective optimization · Recommender systems · Genetic algorithm

Introduction

The big data era enables unprecedented data sharing but also immerses users in information overload, hindering their ability to access the information they need or use their preferred items across various applications [1]. With the emergence of recommendation systems, personalized suggestions can now be provided to users [2, 3]. The goal of recommendation systems is to process large volumes of user-item data by offering the most relevant recommendations to users, thus saving them time and effort in searching for the information they need. By deeply analyzing user preferences and behavior patterns, recommendation systems tailor product, service, or content recommendations to individual users, enhancing the overall user experience [4], with applications spanning e-commerce, entertainment, and social networking [5].

Traditional recommendation systems primarily focus on accuracy as their main objective. However, pursuing accuracy alone often leads to over-recommendation of popular items, resulting in unfair exposure for less popular items. This imbalance disadvantages item providers and negatively impacts the operation of recommendation systems [6]. Enhancing the item exposure fairness can provide opportunities for displaying less popular yet high-quality items, promote diversity in the recommendation ecosystem, and simultaneously meet users' diverse needs, thereby alleviating the problem of information cocoons. Therefore, item exposure fairness has become equally crucial during the recommendation process [7, 8]. Nevertheless, improving non-accuracy metrics like item exposure fairness may compromise recommendation accuracy [9], making the balance between recommendation accuracy and item exposure fairness a significant research challenge [3]. How to skillfully find an equilibrium between accuracy and non-accuracy objectives has become a critical problem requiring urgent res-

✉ Ye Tian
field910921@gmail.com

Liang Chu
2021020031@mail.hfut.edu.cn

1 Hefei University of Technology, Hefei, China

2 Anhui University, Hefei, China

olution. In recent years, significant progress has been made in multi-objective optimization problems. Wang et al. [10] proposed an interactive multi-objective optimization-based MP system. To address expensive multi-objective optimization problems, an adaptive subproblem selection strategy was developed to identify the most promising subproblems [11]. For solving combinatorial multi-objective optimization problems, a single deep reinforcement learning model was proposed [12]. These approaches have effectively addressed various multi-objective optimization problems.

Multi-objective recommendation is an application scenario of multi-objective optimization work, several multi-objective optimization algorithms have been proposed to address the issue of multi-objective recommendation. Zuo introduced a genetic evolutionary algorithm [13]. Wang et al. [14] proposed a recommendation method based on a decomposition based MOEA. Cui et al. developed a novel multi-objective genetic evolutionary algorithm, named PMOEA, to reconcile the conflict between precision and diversity [15]. Jain et al. proposed [16] a multi-parent crossover mechanism called NewCross in their multi-objective recommendation framework, which maintains the order and frequency of parental genes to suggest both popular and diverse items. Although existing multi-objective recommendation methods have achieved certain effectiveness, they often rely on fixed genetic operators, by a large number of experiments or experiences, which is burdensome and could adversely affect the evolutionary effectiveness of the algorithms.

To address this challenge, this paper innovatively proposes a hybrid reinforcement learning-enhanced adaptive evolutionary algorithm (HRL-MOEA), which intelligently adjusts the parameter configurations of genetic operators. This model dynamically adapts the probabilities of genetic operators during evolution, thereby enhancing evolutionary efficiency and convergence speed. The core contributions of this work are summarized as follows:

- We propose a hybrid reinforcement learning-based adaptive multi-objective recommendation algorithm (HRL-MOEA). This algorithm endows the model with self-learning capabilities, enabling adaptive parameter adjustment of evolutionary operators during the optimization process, significantly improving convergence performance.
- By integrating the strengths of SARSA and Q-learning, we design a novel hybrid reinforcement learning strategy: SARSA, as an on-policy algorithm, updates policies by considering actual selected actions, enabling safer exploration of the parameter space and preventing population collapse caused by overly aggressive adjustments, while Q-learning, as an off-policy algorithm, updates policies under the assumption of optimal future actions, thereby accelerating convergence toward optimal parameters. Leveraging these distinct traits, we innovatively formulated a phase-adaptive strategy: during the early evolutionary phase, where high population diversity demands extensive exploration of the solution space to avoid premature convergence, the SARSA strategy is employed to maintain diversity through cautious parameter adjustments; In the late evolutionary phase, as the population converges, the Q-learning strategy is activated to prioritize the exploitation of known effective parameters by fine-tuning crossover probabilities, minimizing random perturbations, and accelerating convergence while avoiding local optima.
- To ensure a smooth transition between the SARSA strategy (adopted in the early evolutionary phase) and the Q-learning strategy (activated in the late phase) rather than abrupt switching that could induce performance fluctuations, we propose a logarithmic complexity-based adaptive threshold to govern the timing of strategy shifts. This threshold is dynamically determined by the action-space complexity and maximum evolutionary generations. Specifically, when the current generation is below the threshold, the SARSA strategy is prioritized to maintain population diversity through cautious exploration; Once the generation exceeds the threshold, the Q-learning strategy takes over to accelerate convergence toward optimal solutions by leveraging refined parameter adjustments.

The subsequent sections of this paper are structured as follows: "Related work" presents the research background and related work, followed by the introduction of a multi-objective recommendation framework and a new multi-objective optimization recommendation algorithm in "The proposed HRL-MOEA". "Experimental studies" discusses the experimental results, and "Conclusion" provides a summary of the paper.

Related work

Recommendation system

In the era of the Internet, people are faced with an overwhelming amount of information, making it exceptionally difficult to sift through and select the information they need. To address this challenge, recommendation systems have emerged [17]. Leveraging advanced technologies such as big data and machine learning, recommendation systems continuously learn users' interests and preferences as they use products, and proactively present items that they may like [18]. This effectively promotes consumption, saves users' time, enhances user experience, optimizes resource alloca-

tion, and creates significant commercial value for service providers and item providers.

The core of a recommendation system lies in its recommendation algorithm. Researchers have proposed various recommendation algorithms, which can be broadly categorized into the following types: collaborative filtering recommendation [19], content-based recommendation [20], social-based recommendation [21], web-based recommendation [22], and hybrid recommendation [23].

With the thriving development of deep learning, neural networks have been widely applied in the field of recommendation systems, significantly enhancing the effectiveness of recommendation algorithms. For example, Oskouei and Hashemzadeh [24] proposed a deep learning-based food recommendation system that comprehensively considers multiple aspects of users and food. This article constructs food feature vector and user feature vector, and combines health standards with personal preferences to accurately recommend suitable food based on the user's health status. In recent years, the application of graph neural networks in the field of recommendation systems has become a new research paradigm [25]. Stergiopoulos et al. [26] utilized graph modeling and deep learning technologies to recommend scientific information, achieving promising results. The recommendation system they proposed, which integrates multiple technologies and algorithms, can efficiently process vast amount of available data in digital academic libraries. In particular, clustering analysis and graph-based methods significantly enhance the performance of the system's artificial neural networks. Wang et al. [27] proposed a new collaborative filtering recommendation algorithm based on graph neural networks (NGCF), which can obtain high-order connectivity representations for each node on the "user-item" bipartite graph. This study presents the first attempt to integrate structural knowledge with the message-passing mechanism in model-based CF, opening up a new research direction. By combining the item knowledge graph with the user-item interaction graph, they establish knowledge-aware connectivities between users and items, thereby revealing the decision-making process behind users' item selections. Subsequently, He et al. [28] further proposed LightGCN, a lightweight and empowered graph convolution network based on NGCF, by eliminating two redundant operations in the model's message passing mechanism, making LightGCN's performance superior to that of NGCF. LightGCN achieves more efficient graph-based collaborative filtering through its minimalist design, resulting in a lighter-weight model with superior performance.

The multi-objective recommendation

The recommendation algorithms discussed above solely prioritize accuracy as the singular evaluation metric. However, as research progresses, researchers have increasingly recognized that such single-objective approaches focused exclusively on accuracy struggle to meet diverse real-world demands. Beyond accuracy, non-accuracy metrics, such as item exposure fairness, is equally critical. This metric ensure that less popular items receive adequate recommendations, which benefits item providers and promotes the healthy operation of recommendation systems [29]. Nevertheless, improving item exposure fairness often conflicts with accuracy optimization, creating a trade-off between accuracy and non-accuracy objectives [3].

To address this challenge, multi-objective recommendation algorithms have emerged. These algorithms can be broadly classified into two categories: one is based on deterministic methods [30–32], which seek optimal solutions through explicit mathematical models, the other is based on stochastic techniques [33, 34], which explore the solution space using randomness and probability to find potential high-quality solutions.

Among the stochastic techniques, multi-objective recommendation algorithms utilizing MOEA have garnered significant attention due to their unique advantages. These algorithms can consider multiple objectives simultaneously and continuously optimize the solution set through selection, crossover, and mutation operations during the evolution process, ultimately recommending a set of optimal solutions. Zuo et al. [13] proposed a framework utilizing NSGA-II to strike a balance between the objectives of diversity and accuracy. Cui et al. [15] introduced a probabilistic multi-objective evolutionary algorithm. Jain et al. [16] proposed a multi-parent crossover mechanism that preserves the order and frequency of parental genes.

Although existing multi-objective recommendation methods have made certain achievements in balancing accuracy and non-accuracy metrics,they often rely on fixed crossover and mutation operators during the evolutionary process. The crossover probability P_c and mutation probability P_m are two critical parameters in evolutionary computation. If P_c and P_m are large, some good individuals could be easily lost, while low values of P_c and P_m may hinder the generation of new promising solutions [35]. To address this issue, this study proposes a hybrid reinforcement learning-based multi-objective evolutionary algorithm (HRL-MOEA). The framework synergistically optimizes genetic operators through adaptive SARSA and Q-learning strategies, thereby enhancing evolutionary efficacy by dynamically balancing exploration-exploitation trade-offs throughout the optimization process.

The mathematical model of multi-objective recommendation is defined as follows,

$$\begin{aligned} \max \quad & F(\mathbf{x}) = \left(f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_s(\mathbf{x})\right)^{\mathrm{T}} \\ \text{s.t.} \quad & \mathbf{x} = (x_1, \ldots, x_n) \in \boldsymbol{\Omega} \end{aligned}$$

where n represents the number of items in a recommendation list, s represents the number of objectives, $\mathbf{\Omega}$ denotes the decision space.

Reinforcement learning

In recent years, research in the field of reinforcement learning has gained increasing momentum, attracting widespread attention [36]. Reinforcement learning is a learning method that aims to maximize cumulative rewards by associating actions with environmental states. It evaluates and adjusts action strategies based on feedback signals from the environment. In this process, the agent continuously interacts with a complex and dynamic environment, continually optimizing its action selections based on immediate environmental feedback, thereby gradually approaching and achieving optimal decision-making.

The model framework of reinforcement learning is shown in Fig. 1. At time t, the agent perceives the current state of the environment S_t and executes an action A_t based on this state. Subsequently, at time t + 1, the environment transitions to a new state S_{t+1} due to the agent's action, and feeds back a reward R_t to the agent. This reward signal prompts the agent to adjust its action selection strategy so that it can take more appropriate action at the next time t + 1. By continuously iterating this process, the agent gradually optimizes its policy, aiming to maximize long-term accumulated rewards.

SARSA and Q-learning are two classic reinforcement learning algorithms, aiming to estimate the Q-values for an optimal policy. The Q-values are updated based on the feedback of state-action pairs and stored in a Q-table, which records the learning experiences of the Agent. Q-learning is an off-policy algorithm that updates under the assumption of optimal action selection to converge more rapidly towards optimal parameters, while SARSA, as an on-policy algorithm, updates Q-values based on the actual actions taken during exploration, enabling safer parameter space exploration by avoiding population collapse caused by aggressive adjustments, with both methods employing distinct Q-value update functions [37].

The Q-value update function for SARSA is as follows,

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha(R_t + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)). \quad (1)$$

The Q-value update function for Q-learning is as follows,

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha(R_t + \gamma \max_{a_{t+1}}(Q(s_{t+1}, a_{t+1})) - Q(s_t, a_t)) \quad (2)$$

where $Q(s_t, a_t)$ is the Q value of taking action a_t in current state s_t, R_t is the reward value after the execution of action a_t at state s_t, α represents the learning rate, and γ is a discount rate to control the convergence. The expression $\max_{a_{t+1}} Q(s_{t+1}, a_{t+1})$ represents the maximum Q-value among all possible actions in the next state s_{t+1}, independent of the actual action taken. SARSA updates Q-values using the actual subsequent action a_{t+1} to reflect the exploration process of the current policy, while Q-learning directly assumes the selection of the optimal action and ignores the actual exploration path.

The proposed HRL-MOEA

Algorithm framework

In multi-objective recommendation algorithms, genetic operators play a critical role. Inappropriate genetic operators may hinder the generation of high-quality individuals during the evolutionary process or even lead to the loss of excellent individuals. Traditional multi-objective recommendation algorithms typically rely on a large number of experiments or experiences to determine fixed genetic operators, which is burdensome.

To address this challenge, this paper proposes an innovative hybrid reinforcement learning-based adaptive multi-objective recommendation algorithm, named HRL-MOEA. The algorithm integrates characteristics from both SARSA and Q-learning reinforcement learning algorithms, allowing the model to appropriately adjust genetic operators while learning from the positive influence of previous generations on subsequent evolutionary processes.

The framework of the HRL-MOEA algorithm is shown in Fig. 2, encompassing key stages such as candidate set generation, population initialization, crossover operation, mutation operation, and elitist strategy. We obtain a list of candidate sets through LightGCN [28], the population initialization adopts an extremum-guided method, introducing population with high LightGCN scores and population with low exposure at this process, the crossover operation is similar to [16], and the mutation operation employs single-point mutation.

In each mutation operation, a gene is randomly selected from the individual. When the mutation probability condition is met in this instance, the mutation operation is executed. When a mutation occurs, it must be ensured that the mutated gene does not duplicate any existing genes within the individual.

In the elitism strategy process, a specified number of generated offspring populations are merged with the parent population. The combined population is then sorted and stratified using the fast non-dominated sorting method. Subsequently, the optimal solution set is selected hierarchically from the multiple layers. If the number of selected optimal solutions exceeds the size of the parent population, the

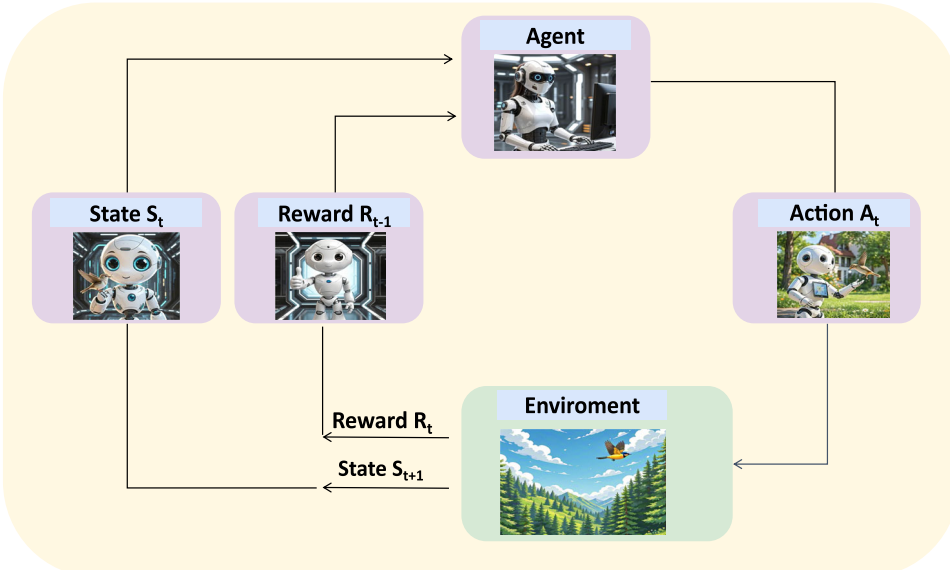


Fig. 1 The framework of reinforcement learning

crowding distance of the solutions in the last selected layer is computed. These solutions are then sorted based on their crowding distance, and those with lower crowding distances are eliminated to ensure the offspring population size equals that of the parent population.

Every k generations, the model updates the Q-table through reinforcement learning strategy selection and performs action selection. By employing reinforcement learning, the agent can intelligently learn and adjust its crossover strategy and mutation strategy, particularly by dynamically optimizing the crossover probability P_c and the mutation probability P_m, based on the current state and the feedback from the reward.

Construct learning module

SARSA, as an on-policy algorithm, updates Q-values based on the actually selected actions, enabling safer exploration of the parameter space and avoiding population collapse caused by aggressive adjustments. In contrast, Q-learning, an off-policy algorithm, updates Q-values under the assumption of selecting optimal actions, allowing faster convergence toward near-optimal parameters. By integrating the strengths of both SARSA and Q-learning, we designed a hybrid reinforcement learning strategy, the architecture of the learning module is illustrated in Fig. 3. During the early stages of the evolutionary process, where high population diversity necessitates broad exploration of the solution space, the action selection follows the SARSA strategy. As the population converges in later stages, requiring fine-tuned adjustments to crossover probabilities to accelerate convergence and evade local optima, the Q-learning strategy governs action selection. In the early stages of evolution, the SARSA strategy is adopted, with Q-values updated through Eq. (1) and recorded in the Q-table, In the later stages of evolution, the Q-learning strategy is implemented, with Q-values updated via Eq. (2) and also stored in the Q-table. The transition formulas are as follows:

$$RL = \begin{cases} SARSA, & t < T \\ Q - learning, & t >= T \end{cases} \tag{3}$$

where t represents the current evolutionary generation, and T denotes the threshold number for strategy transition.

To ensure a smooth transition between these phases and avoid abrupt switches that might induce performance fluctuations, we propose a logarithmic complexity-based adaptive threshold to control the timing of strategy switching. The threshold formula is defined as follows,

$$T = \lambda \cdot \frac{\ln(S \cdot A + 1)}{\ln(2)} \cdot \sqrt{g_{max}} \tag{4}$$

where λ is the adjustment coefficient that controls the transition speed, and in this study, the value of λ is set to 0.5, S represents the number of state spaces, A denotes the number of actions, and g_{max} refers to the predefined maximum number of evolutionary generations. When the evolutionary generation is less than the threshold T, the SARSA strategy is adopted; when it exceeds T, the system switches to the Q-learning strategy.

Construct objective function

In this paper, two conflicting objectives are considered. The first one measures the accuracy of recommendations, ratings can be estimated by recommendation models [38], and our

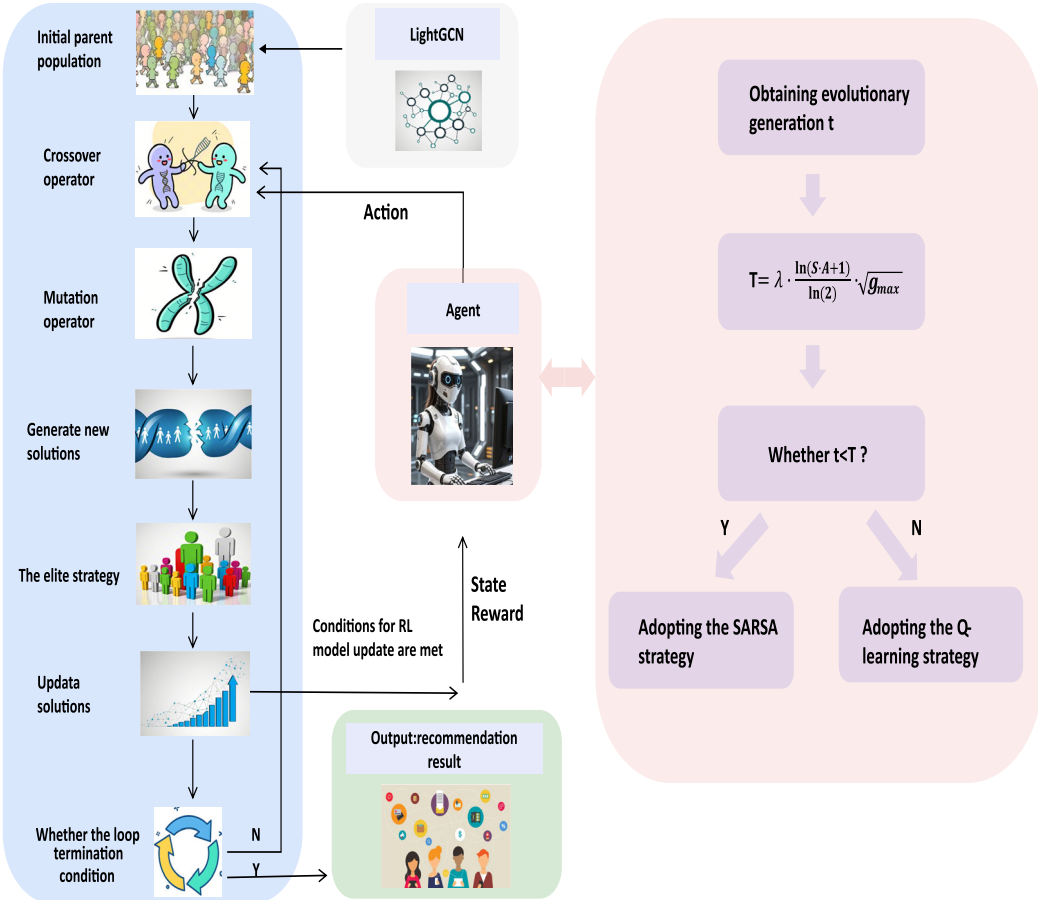
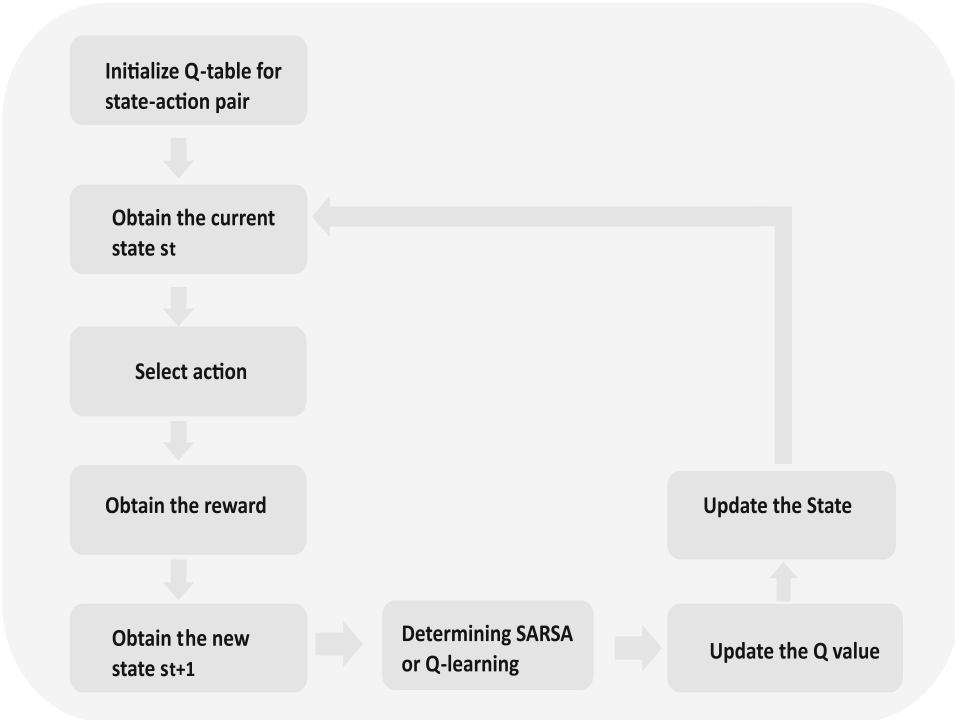


Fig. 2 The framework of HRL-MOEA

Fig. 3 The architecture of the learning module



objective function of accuracy is based on the evaluated rating of items derived from the LightGCN. The objection function of accuracy is defined as follows,

$$f(acc) = \frac{1}{|\mathbf{x}|} \sum_{i \in \mathbf{x}} \hat{\mathbf{R}}_{ui} \tag{5}$$

where $\hat{\mathbf{R}}_{ui}$ denotes the predicted accuracy score of a user u for an item i, $\mathbf{x}$ denotes the recommendation list for user u.

In modern recommender systems, the item exposure fairness has become a critical consideration. In this study, items are ranked based on their interaction frequency with users and categorized into popular and unpopular groups. Specifically, we sort items in descending order according to their user interaction frequency. Similar to [39], items are divided into two groups: the top 20% with the highest interaction counts are classified as the popular group, while the remaining 80% form the unpopular group.

In a recommendation list, items positioned higher receive greater exposure opportunities. To quantify the item exposure fairness, we measure the prominence of unpopular group items within the recommendation list. Specifically, the DCG and IDCG metrics are employed, as defined in Eq. (6) and Eq. (7), respectively.

$$DCG = \sum_{i}^{K} \frac{1}{log_2(i+1)} \tag{6}$$

$$IDCG = \sum_{i=1}^{K} \frac{1}{log_2(i+1)}. \tag{7}$$

In the DCG formula, i represents the position of an unpopular group item within the recommendation list. The IDCG metric is utilized to normalize the DCG score, where i in the IDCG formula denotes the position of any item (regardless of popularity) in the recommendation list. The item exposure fairness metric for the recommendation list is formulated as follows,

$$f(fair) = \frac{u_{(s)}}{u_{(L)}} \cdot \frac{u_{(DCG)}}{u_{(IDCG)}} \tag{8}$$

where $u_{(s)}$ denotes the number of items belonging to the unpopular item group that appears in the recommended list, whereas $u_{(L)}$ represents the overall count of items present in the entire recommended list.

The two objective functions aim to improve the accuracy and item exposure fairness of recommendations, respectively. It is evident that there is a certain conflict between the accuracy metric and the item exposure fairness metric. Therefore, we adopt a multi-objective optimization algorithm to properly address this conflict. The objective functions optimized in this paper are summarized as follows:

$$\begin{cases} Max & f(acc) \\ Max & f(fair). \end{cases} \tag{9}$$

State set

In the course of designing the algorithm presented in this study,, the environment state is closely related to the fitness of the population. The environment state is specifically composed of changes in the fitness values of the population and the diversity of the population. The calculation formula for the fitness of the population is given in Eq. (10), and the formula for population diversity is given in Eq. (11). Equation (12) represents the overall state value, obtained through a weighted combination of Eqs. (10) and (11):

$$State_1 = \frac{\sum_{k=1}^{M} f_{(x_k^1)}}{\sum_{k=1}^{M} f_{(x_k^t)}} \tag{10}$$

$$State_2 = \frac{\sum_{k=1}^{M} \left| f_{(x_k^t)} - \frac{\sum_{k=1}^{M} f_{(x_k^t)}}{M} \right|}{\sum_{k=1}^{M} \left| f_{(x_k^1)} - \frac{\sum_{k=1}^{M} f_{(x_k^1)}}{M} \right|} \tag{11}$$

$$State = \epsilon \cdot State_1 + (1-\epsilon) \cdot State_2 \tag{12}$$

where $f(x_k^t)$ represents the fitness of the k-th individual in the t-th generation, $f(x_k^1)$ represents the fitness of the k-th individual in the first generation, ϵ is a weight parameter that can determine the importance of two factors and the range of ϵ is between 0 and 1. Both fitness of the population and population diversity are critical for enhancing the quality of the population. In this study, the ϵ was set to 0.5.

The value of state ranges from 0 to 1, based on the value of state, the state set is divided into 20 states, and the interval value of state is set as 0.05.

Action set

In multi-objective recommendation algorithms, crossover and mutation operations play crucial roles, where the crossover probability P_c and the mutation probability P_m are the core parameters that directly determine algorithmic performance. The inappropriate probability values may hinder the generation of superior individuals during evolution and potentially lead to the loss of excellent individuals. For each state, the agent will take different actions to obtain an appropriate P_c and P_m, with the action set containing 10 actions.

For the crossover probability P_c, its commonly used values range from 0.5 to 0.8, and the interval value between each action is 0.03, for instance, when the action is set to a_1,

$P_c \in [0.5, 0.53)$, and a value is randomly sampled within this interval; when action a_2 is chosen, $P_c \in [0.53, 0.56)$. The same methodology applies to the mutation probability P_m, which operates within the range from 0.4 to 0.7 and follows the 0.03 interval. For example, when the action is set to a_1, $P_m \in [0.4, 0.43)$, and a random value within this range will be chosen.

Reward

During the design process of the algorithm proposed in this study, the reward is designed based on the individual's best fitness, the population's overall fitness, and population diversity. The reward for the best individual fitness is shown in Eq. (13), the overall fitness reward is defined in Eq. (14), and the population diversity reward is formulated in Eq. (15). Equation (16) represents the total reward value obtained through the weighted combination of Eqs. (13), (14), and (15). Here, β_1, β_2, and β_3 are weight parameters that determine the importance of the three factors, with the sum of these weights equal to 1. In this study, population fitness and population diversity reflect the state of the entire population, which benefits the improvement of overall population quality and facilitates the acquisition of superior individuals. Consequently, the weights for population fitness and population diversity are set relatively larger than that for individual fitness. Specifically, the value of β_1 is 0.3, whileβ_2 and β_3 are both set to 0.35.

$$Reward_1 = \frac{maxf_{(x_k^t)} - maxf_{(x_k^{t-1})}}{maxf_{(x_k^{t-1})}} \tag{13}$$

$$Reward_2 = \frac{\sum_{k=1}^{M} f_{(x_k^t)} - f_{(x_k^{t-1})}}{\sum_{k=1}^{M} f_{(x_k^{t-1})}} \tag{14}$$

$$Reward_3 = \frac{\sum_{k=1}^{M} \left| f_{(x_k^{t-1})} - \frac{\sum_{k=1}^{M} f_{(x_k^{t-1})}}{M} \right|}{\sum_{k=1}^{M} \left| f_{(x_k^{t})} - \frac{\sum_{k=1}^{M} f_{(x_k^{t})}}{M} \right|} \tag{15}$$

$$Reward = \beta_1 \cdot Reward_1 + \beta_2 \cdot Reward_2 + \beta_3 \cdot Reward_3. \tag{16}$$

The process of HRL-MOEA

Based on above work, the execution flow of the entire HRL-MOEA is described in Algorithm 1.

Algorithm 1: Procedure of HRL-MOEA

Input: $\mathbf{R}$(user-item interaction matrix),n(number of solutions),g_{max}(maximum number of generations)
Output: P_{final} (set of recommendation lists)

```
Initialize the RL:Q value table, state set, action set;
R̂ ← LightGCN(R) // Estimate the ratings of users on all the unrated items based on R;
P ← Initialization(R̂,n) // Initialize the parent population;
Choose a random action ;
while terminal condition is not fulfilled do
    for each generation population P_i do
        P'_i ← Crossover(P_i) // Perform crossover operation;
        P''_i ← Mutation(P'_i) // Perform mutation operation;
        P'''_i ← P''_i + P_i ;
        Evaluate each solution in P'''_i by objective functions;
        P_i ← Non-dominated sorting(P'''_i, n) // Update the parent population via elitism strategy;
        Compute the threshold T by Eq.(4);
        if generation t==1 then
            obtain the fitness values and population diversity values of the first generation population by Eqs.(10)(11).
        end
        if conditions for RL model update are met then
            Obtain the state by Eq.(12);
            Obtain the reward by Eq.(16) ;
            if generation t < T then
                Adopt the SARSA strategy
            else
                Adopt the Q-learning strategy
            end
            Update the Q-table, and select the action
        end
    end
end
P_final ← FinalSelection(P_i);
return P_final;
```

Experimental studies

Datasets

Two well-known and freely available datasets Movielens100K and Movielens1M are used to evaluate the performance and effectiveness of our proposed algorithm. Movielens100K is a renowned public dataset for recommendation research, containing 100,000 ratings from 943 users on 1682 movies, with each user having rated at least 20 movies. Movielens1M contains 100 million ratings from 6000 users on 4000 movies. The dataset is split into a training set and a test set, with 80% allocated to the training set and the remaining 20% to the test set.

Table 1 Results of HV

Dataset	Qualities	MOEA-Probs	MORS	PMOEA	NewCrossPMOEA	HRL-MOEA
Movielens100k	Mean	0.2613(+)	0.2636(+)	0.2571(+)	0.2668(+)	**0.2875**
	std	2.33E−03	3.24E−03	1.89E−03	3.71E−03	2.94E−03
Movielens1M	Mean	0.2966(+)	0.2991(+)	0.2894(+)	0.3019(+)	**0.3297**
	std	2.77E−03	3.74E−03	3.08E−03	4.87E−03	3.43E−03
+/−/≈		2/0/0	2/0/0	2/0/0	2/0/0	

Results of comparing HV(the larger, the better) of HRL-MOEA with four comparison algorithms, where "+", "−" and "≈" indicate that the performance is significantly better, significantly worse and statistically similar to that of HRL-MOEA, respectively

Table 2 Results of spread

Dataset	Qualities	MOEA-Probs	MORS	PMOEA	NewCrossPMOEA	HRL-MOEA
Movielens100k	Mean	0.8594(−)	0.8725(−)	0.9331(−)	0.9048(−)	**0.8376**
	std	5.22E−03	4.91E−03	5.84E−03	6.37E−03	5.56E−03
Movielens1M	Mean	0.9157(−)	0.9239(−)	0.9709(−)	0.9511(−)	**0.8728**
	std	5.48E−03	5.94E−03	7.36E−03	6.47E−03	6.16E−03
+/−/≈		0/2/0	0/2/0	0/2/0	0/2/0	

Results of comparing Spread(the smaller, the better) of HRL-MOEA with four comparison algorithms, where "−", "+" and "≈" indicate that the performance is significantly better, significantly worse and statistically similar to that of HRL-MOEA, respectively

Table 3 Results of HV

Dataset	Qualities	HRL-MOEA(-CRO)	HRL-MOEA(-MUT)	HRL-MOEA
Movielens100k	Mean	0.2816(+)	0.2799(+)	**0.2875**
	std	3.45E−03	2.67E−03	2.94E−03
Movielens1M	Mean	0.3223(+)	0.3197(+)	**0.3297**
	std	3.15E−03	3.71E−03	3.43E−03
+/−/≈		2/0/0	2/0/0	

Results of comparing HV(the larger, the better) of HRL-MOEA with two variants, where "+", "−" and "≈" indicate that the performance is significantly better, significantly worse and statistically similar to that of HRL-MOEA, respectively

Evaluation metrics

Hypervolume (HV) [40] is a metric used to evaluate the convergence of multi-objective optimization, widely applied in the assessment of multi-objective evolutionary algorithms. Since the optimization problem in this paper is a multi-objective maximization problem, we select (0,0) as the reference point for HV calculation. A larger HV value indicates better quality of the solutions.

Spread [41] is also a key performance metric in the field of multi-objective optimization for evaluating the distribution characteristics of solution sets generated in the objective space. A smaller Spread value indicates better quality of the solutions.

Table 4 Results of spread

Dataset	Qualities	HRL-MOEA(-CRO)	HRL-MOEA(-MUT)	HRL-MOEA
Movielens100k	Mean	0.8486(−)	0.8521(−)	**0.8376**
	std	5.36E−03	5.86E−03	5.56E−03
Movielens1M	Mean	0.8923(−)	0.9013(−)	**0.8728**
	std	6.38E−03	6.71E−03	6.16E−03
+/−/≈		0/2/0	0/2/0	

Results of comparing Spread(the smaller, the better) of HRL-MOEA with two variants, where "−", "+" and "≈" indicate that the performance is significantly better, significantly worse and statistically similar to that of HRL-MOEA, respectively

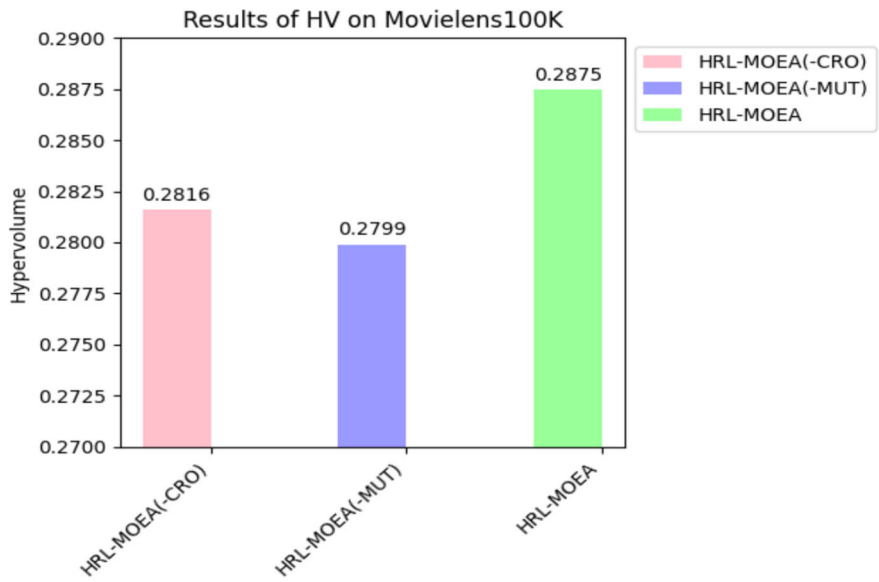


Fig. 4 Results of HV on Movielens100K

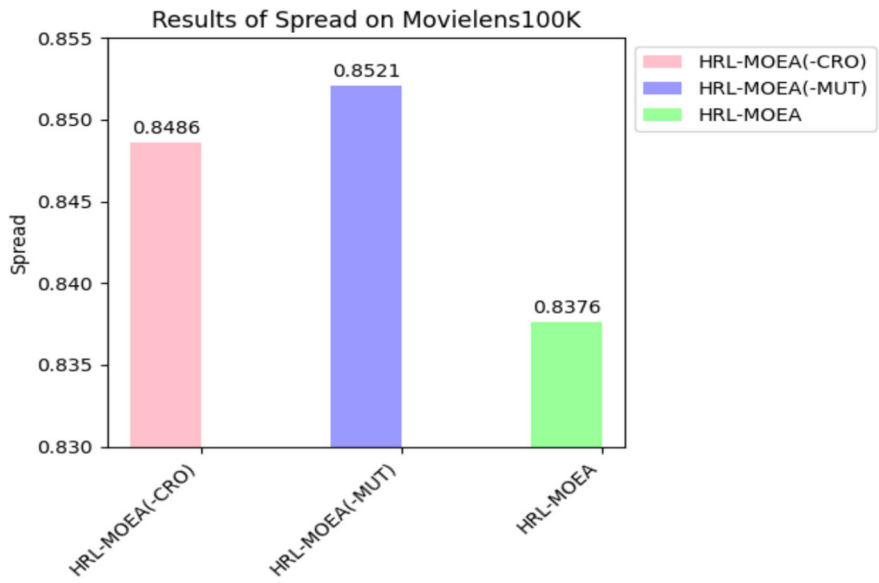


Fig. 5 Results of spread on Movielens100K

Comparative methods

To validate the effectiveness of the proposed algorithm, the proposed method HRL-MOEA is compared with the following four MOEA-based recommendation algorithms:

- NewCrossPMOEA [16] is a recommendation method, which preserves the order and the frequency in the parents' genes to bring good objectivity in the trade-off of recommending popular and diverse items in the recommendation list.
- PMOEA [15] is a recommendation method based on a probabilistic MOEA, which searches for top-K items for all users simultaneously by maximizing accuracy and diversity.
- MORS [14] is a recommendation method based on a decomposition based MOEA, which searches for top-K items for each user.
- MOEA-Probs [13] is a recommendation method based on NSGA-II.

Parameter settings

The number of solutions is set to 30, the length of the recommended list is set to 10. The number of generations for the evolutionary computation is set to 400. For the comparative algorithms, the crossover probability is set to 0.6, and the mutation probability is set to 0.5.

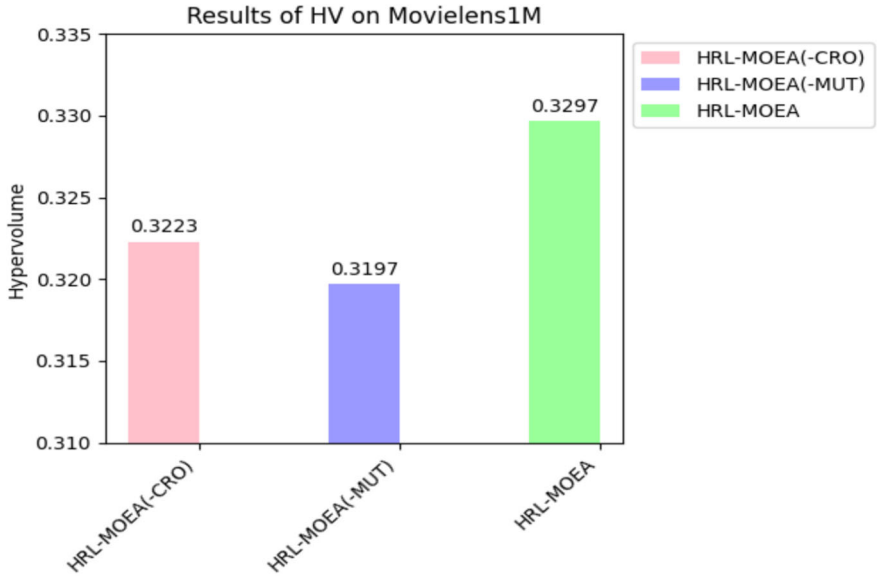


Fig. 6 Results of HV on Movielens1M

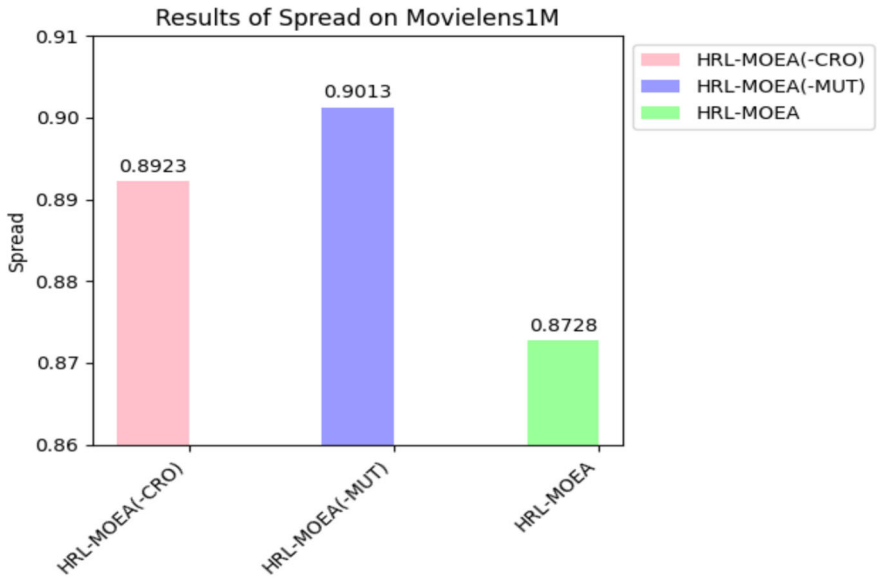


Fig. 7 Results of spread on Movielens1M

Experimental results

In this section, we first validate the effectiveness of the proposed HRL-MOEA by comparing its solutions with baseline algorithms, and then verify the efficacy of the proposed strategies through comparisons with variants of HRL-MOEA.

Comparison results between HRL-MOEA and baselines

In this study, we evaluated the effectiveness of the proposed algorithm through experiments conducted on two datasets (Movielens100k and Movielens1M). The evaluation metrics for individual users within the datasets were computed, followed by a comparative analysis using averaged results across all users.

In this experiment, each algorithm was independently executed 10 times, with all results averaged to derive the final outcomes. The experimental results for the HV metric are shown in Table 1, and those for the Spread metric in Table 2. Mathematical statistics were also performed using Wilcoxon rank sum test with a significance level of 0.05 and the optimal values in the tables are marked in bold, where "+", "−" and "≈" indicate the results of the Wilcoxon significance test obtained by HRL-MOEA compared to the baseline. For the HV metric, a higher number of "+" symbols indicates better performance of the proposed method, whereas for the Spread metric, a higher number of "−" symbols reflects superior performance of the method.

From the experimental results, our proposed HRL-MOEA demonstrates superior performance over comparative algo-

Table 5 Results of HV

Dataset	Qualities	HRL-MOEA(SARSA)	HRL-MOEA(Q-learning)	HRL-MOEA
Movielens100k	Mean	0.2845(+)	0.2836(+)	**0.2875**
	std	3.13E−03	3.61E−03	2.94E−03
Movielens1M	Mean	0.3255(+)	0.3241(+)	**0.3297**
	std	3.67E−03	3.94E−03	3.43E−03
+/−/≈		2/0/0	2/0/0	

Results of comparing HV(the larger, the better) of HRL-MOEA with two variants, where "+", "−" and "≈" indicate that the performance is significantly better, significantly worse and statistically similar to that of HRL-MOEA, respectively

Table 6 Results of spread

Dataset	Qualities	HRL-MOEA(SARSA)	HRL-MOEA(Q-learning)	HRL-MOEA
Movielens100k	Mean	0.8409(−)	0.8451(−)	**0.8376**
	std	6.68E−03	5.95E−03	5.56E−03
Movielens1M	Mean	0.8834(−)	0.8872(−)	**0.8728**
	std	6.49E−03	6.88E−03	6.16E−03
+/−/≈		0/2/0	0/2/0	

Results of comparing Spread(the smaller, the better) of HRL-MOEA with two variants, where "−", "+" and "≈" indicate that the performance is significantly better, significantly worse and statistically similar to that of HRL-MOEA, respectively

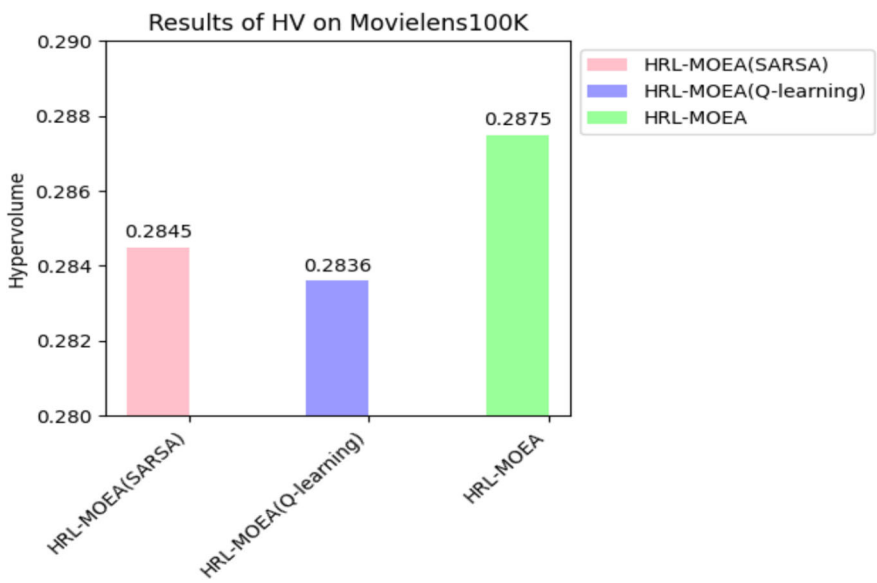


Fig. 8 Results of HV on Movielens100K

rithms in terms of the Hypervolume and Spread indicators. This is attributed to our reinforcement learning-integrated strategy for selecting crossover and mutation operators. In this reinforcement learning framework, we utilize the population's fitness and population diversity as reward for the model. This design enables the model to preferentially choose operators that benefit both fitness optimization and diversity preservation during the evolutionary process. Such a strategic selection mechanism effectively enhances the algorithm"s evolutionary efficacy, thereby achieving improved optimization outcomes.

Effectiveness of the proposed strategies

To validate the effectiveness of the proposed strategy, we conducted experimental comparisons between HRL-MOEA and four variants of it. In this experiment, each algorithm was independently executed 10 times, with all results averaged to derive the final outcomes. Specifically, we first selectively removed the reinforcement learning module from the crossover and mutation operations of HRL-MOEA. The experimental results are summarized in Tables 3 and 4. To enhance the visualization effects, we have presented the results in Figs. 4, 5, 6, and 7.

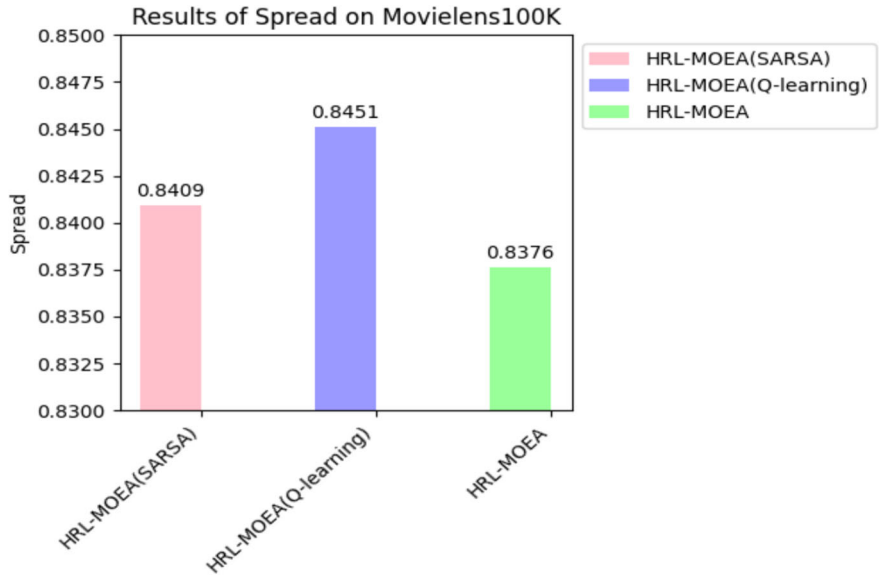


Fig. 9 Results of spread on Movielens100K

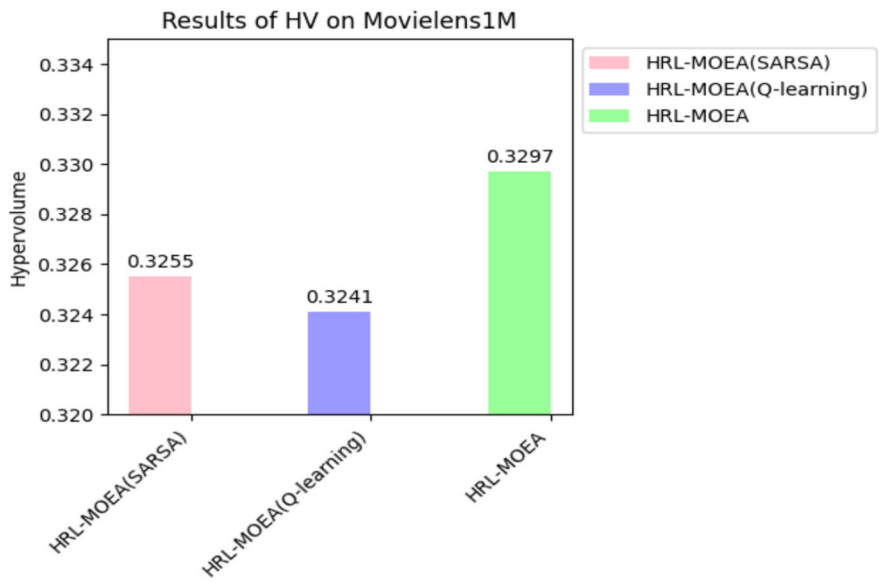


Fig. 10 Results of HV on Movielens1M

In the tables, HRL-MOEA(-CRO) is the same algorithm with HRL-MOEA except the reinforcement learning-integrated crossover module is removed, while HRL-MOES(-MUT) is signifies the variant where the reinforcement learning-enhanced mutation module is eliminated. From the experimental results, HRL-MOEA outperforms both HRL-MOEA(-MUT) and HRL-MOEA(-CRO). This superiority is primarily attributed to our proposed algorithm's reinforcement learning-integrated strategy, which adaptively selects operators that enhance population fitness and diversity, thereby improving evolutionary effectiveness. HRL-MOEA(-MUT) exhibits inferior metrics compared to HRL-MOEA(-CRO), indicating that the reinforcement learning-driven mutation mechanism contributes more significantly to performance than its crossover counterpart.

To verify the effectiveness of the hybrid reinforcement learning strategy, we conducted experiments using the SARSA and Q-learning reinforcement learning strategies independently. The experimental results are shown in Tables 5 and 6. To enhance the visualization effects, we have presented the results in Figs. 8, 9, 10, and 11.

In the tables, HRL-MOEA(SARSA) denotes the variant where only the SARSA reinforcement learning strategy is applied to optimize crossover and mutation operations, while HRL-MOEA(Q-learning) represents the version exclusively utilizing the Q-learning strategy. As evidenced by the experimental results, the proposed hybrid reinforcement learning model HRL-MOEA outperforms both HRL-MOEA(SARSA) and HRL-MOEA(Q-learning). This superiority stems from the dynamic integration mecha-

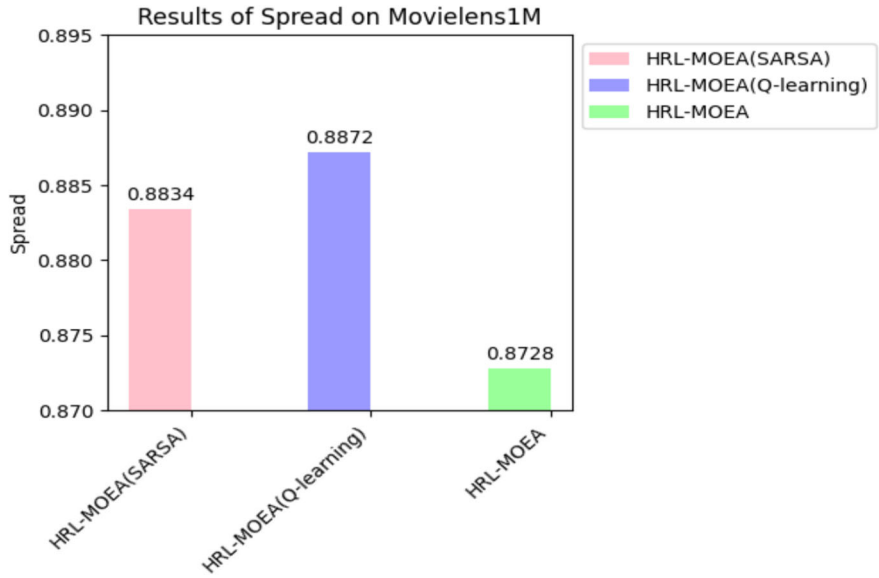


Fig. 11 Results of spread on Movielens1M

nism: the hybrid model initially employs the conservative SARSA strategy to facilitate broad exploration of the solution space during early evolutionary stages, then strategically switches to the Q-learning strategy in later phases to accelerate convergence while mitigating unnecessary oscillations. Such synergistic coordination enables HRL-MOEA to achieve enhanced evolutionary effectiveness through balanced exploration-exploitation trade-offs.

Discussion

In comparative experiments against baselines, the proposed HRL-MOEA demonstrates superior performance, which is primarily attributed to its integration of a reinforcement learning strategy. By leveraging population fitness and population diversity as reward, the algorithm dynamically optimizes the selection of crossover and mutation operators during the evolutionary process, thereby enhancing overall evolutionary efficacy. The experimental results from HRL-MOEA variants provide further validation of our hybrid reinforcement learning strategy's efficacy, whereby the initial adoption of the SARSA strategy facilitates extensive exploration of the solution space, while the subsequent strategic transition to Q-learning accelerates algorithm convergence and mitigates oscillations. This synergistic orchestration dynamically balances exploration-exploitation interactions through adaptive policy switching, endowing HRL-MOEA with heightened optimization efficacy in evolutionary search processes.

Conclusion

To address the challenges in multi-objective recommendation and enhance algorithmic performance, this study proposes HRL-MOEA, a multi-objective recommendation algorithm incorporating hybrid reinforcement learning strategies. The framework integrates the complementary strengths of SARSA and Q-learning through a phase-aware hybridization mechanism: during the initial exploration phase, SARSA's conservative policy adaptively selects crossover and mutation operators to thoroughly probe the solution space, while strategically transitioning to Q-learning's aggressive policy in later stages to accelerate convergence by optimizing exploitation patterns. This dynamic policy orchestration enhances evolutionary performance through context-aware balance of exploration-exploitation dynamics. Experimental results demonstrate that HRL-MOEA achieves superior performance compared to baselines and variants of HRL-MOEA across critical evaluation metrics.

Author Contributions All authors contributed to the study conception and design. The experiment and data analysis were performed by Liang Chu. The first draft of the manuscript was written by Liang Chu. Ye Tian and Liang Chu revised the manuscript. All authors supervised the development of the manuscript.

Data availability Statement Data will be made available on request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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